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# Ctta: a novel chain-of-thought transfer adversarial attacks framework for large language models

Xinxin Yue[1,2,3], Zhiyong Zhang[1,2,3*], Junchang Jing[4] and Weiguo Wang[5]

**Abstract**

Recent studies have indicated that large language models (LLMs) remain susceptible to adversarial attacks, despite enhanced robustness through the chain-of-thought (CoT) capability. However, this capability also introduces the potential for more covert and effective adversarial attack methods. This paper proposes a CoT Transfer Adversarial attack framework (CTTA) for general LLMs. Initially, we utilize a pre-trained model based on the transformer architecture and fine-tune it on various tasks to serve as a surrogate model. Subsequently, different levels of adversarial attack algorithms are utilized, and the generated adversarial samples are used as transfer samples. A thought chain-based adversarial transfer attack framework is constructed using transfer samples and thought chain techniques. Finally, various indicators are utilized to assess the performance of the general LLMs in response to this attack. The results demonstrate that the attack framework surpasses current state-of-the-art research. Numerous experiments on LLMs with varying performance and parameter sizes have validated the effectiveness, stability, and generalizability of this attack. The model's error response and the superiority of this attack are thoroughly examined using attention by gradient technology, confirming the security threats posed by LLMs when leveraging CoT capability. This has significant implications for enhancing the security and robustness of LLMs.

**Keywords** Adversarial attack, Chain-of-thought, Large language models, Transfer-based attack

## Introduction

Due to the rapid development of artificial intelligence technology, human society has entered an era in which technology and philosophy are intersections, and LLMs are at the center of the intersection (Shanahan 2024). LLMs usually have a massive parameter scale and complex network structure. They are built on the Transformer architecture (Vaswani et al. 2017) and have the training paradigm of "pre-training + fine-tuning". At present, the research of LLMs has made remarkable progress, and its parameter magnitude has crossed the threshold of 100 billion (Brown et al. 2020), with excellent learning, question and answer, reasoning, math, code generation, text generation, machine translation, and other capabilities,which have triggered a series of changes in human-computer interaction, scientific research, content creation, and other fields (Chang et al. 2024). With the

*Correspondence:
Zhiyong Zhang
xidianzzy@126.com
[1] Information Engineering College, Henan University of Science and Technology, Luoyang 471023, China
[2] Henan International Joint Laboratory of Cyberspace Security Applications, Henan University of Science and Technology, Luoyang 471023, China
[3] Henan Intelligent Manufacturing Big Data Development Innovation Laboratory, Henan University of Science and Technology, Luoyang 471023, China
[4] College of Computer and Information Engineering, Henan Normal University, Xinxiang 453007, China
[5] Henan Representative Office, New H3C Technologies Co., Ltd, Zhengzhou 450040, China

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expansion of model size, the ability of LLMs has a qualitative leap, and there is an ability that does not exist in small models but is generated in LLMs, called emergent capability (Zoph et al. 2022), such as in context learning (ICL), Chain of thought (CoT), etc. ICL enables LLMs to mimic existing examples to solve new problems by adding input–output samples to the prompts; CoT adds intermediate reasoning steps to the prompts. Therefore, the emergent capability in downstream task scenarios show strong application potential.

CoT enables the model to show strong logical reasoning capability when dealing with complex problems (Wei et al. 2022). At the same time, this capability also provides an opportunity for attackers, who can use the complexity of the CoT to construct more elaborate and hidden adversarial samples. These samples can disrupt the model's reasoning process and cause the model to make incorrect decisions at critical points. The transferability of adversarial examples further aggravates the severity of this issue (Li et al. 2024). Due to the similarity and interoperability between LLMs, attackers can transfer the adversarial examples generated on one model to another to achieve cross-model attacks. This is due to the need to consider the security of multiple models simultaneously. Applying LLMs in critical domains such as finance and healthcare is pivotal to the accuracy of decision-making. However, if the attacker can successfully use the adversarial samples to carry out a transfer-based attack to interfere with the decision-making process of these models, it may result in incalculable economic losses or security risks.For instance, when a medical large language model is subjected to adversarial attacks, it may provide incorrect recommendations in diagnostic systems, leading to erroneous or delayed treatments, which directly threaten patient health. Similarly, adversarial attacks on financial large language models can result in financial assistants offering inaccurate investment advice, causing users to incur economic losses and affecting the stability of financial markets.

To address the above problems, this paper utilizes different perturbation levels of adversarial attacks on different General LLMs, and different types of tasks, targeting the key data parts in the prompt samples and combining the CoT to construct the adversarial transfer prompt samples against different General LLMs. Moreover, this paper comprehensively evaluates the performance differences of different LLMs when facing adversarial prompt samples by comparing them with baseline attack methods and discussing the effectiveness and universal applicability of CoT-based transfer adversarial attacks on LLMs. Eventually, based on the information about the attention distribution of the LLMs, we can further explore the reasons why the LLMs are misled under the strong logical reasoning capability of CoT, and find the adversarial weakness of the CoT of the LLMs, to provide support for the enhancement of the robustness of the LLMs, and help to realize the LLMs' active security. This paper will provide support for improving the robustness of LLMs and help realize LLMs' active safety.Hence, the contributions of this paper are as follows.

- To maintain the structural similarity between the substitute model and the target model, this paper uses a pre-trained model based on Transformer and obtains the fine-tuning version of this model on different tasks as the substitute model. At the same time, the OpenAttack framework is used to get different perturbation levels.
- In the process of constructing transfer adversarial samples, this article extracts the optimal task instructions in the PromptBench framework to ensure the initial performance of the LLMs. To effectively stimulate the CoT capability, the CoT technology is utilized to fuse the optimal CoT trigger and the LLMs reasoning process.
- This study focuses on LLMs with varying performance and parameter scales as the target models. Single perturbation level and hybrid perturbation levels' adversarial attacks are implemented, while DAC, ADAC, APDR, and ASR are used as evaluation indexes to demonstrate the validity and general applicability of CTTA.
- CosSim, PPL, and GErr are used to assess the quality of CTTA-Prompt samples, and the Attention by Gradient technique visualizes the distribution of Attention to discuss the causes of the model's erroneous responses and the advantages of incorporating the CoT technique for adversarial attacks.

## Related work

### Textual adversarial attacks

An adversarial attack deceives the victim model by adding perturbations to the original samples to cause the model to produce false outputs. An adversarial attack originated in the field of computer vision, and many classical adversarial attack algorithms have been proposed, such as C&W Attack, UAP, FGSM, GAN, etc (Carlini and Wagner 2017; Moosavi-Dezfooli et al. 2017; Zhang et al. 2023; Zhao et al. 2018). Due to the discreteness of text, the image adversarial attack algorithm cannot be directly used for text adversarial attacks. However, a series of adversarial attack algorithms are derived from the field of text adversarial attack. These attack algorithms are classified into character-level, word-level, sentence-level, and multi-level. Character-level attacks mainly perturb the

characters in the original input, including the addition, deletion, replacement, and exchange of characters. DeepWordBug (Gao et al. 2018) algorithm uses the character-level modification algorithm to attack important words, and the TextBugger (Li et al. 2019) algorithm combines character-level and word-level attacks. It implements both black-box and white-box attacks. Word-level attacks mainly scramble the input text by word substitution and have a higher quality of adversarial samples compared to word-level attacks. The BERT-Attack (Li et al. 2020) algorithm performs word substitution based on the ordering of importance of words obtained from the masking of words. Compared with the same level text attack model TextFooler (Jin et al. 2020), BERT-Attack significantly improves the generation efficiency with the help of the masked language model of BERT (Kenton and Toutanova 2019) and ensures the correctness and continuity of syntax and semantics; sentence-level perturbation treats the whole sentence as the object of perturbation and usually realizes the attack by using rephrasing, re-decoding after encoding, and adding irrelevant sentences, and its perturbation. The result often leads to a large gap between the adversarial samples and the original input, which makes it difficult to ensure the quality of the adversarial samples and their validity. In this study, the StressTest (Naik et al. 2018) and CheckList (Bhatt et al. 2021) algorithms integrated into the attack framework realize the adversarial attack by adding irrelevant content at the end of the prompt data.

A transfer-based adversarial attack occurs when the attacker generates adversarial samples by using the known model; these samples can mislead the model to make wrong judgments under the condition that the human eye is difficult to detect. The attacker then applies these adversarial samples to other models with different structures and parameters. To attack unknown models, the target model and the substitute model usually have similar model architectures (Li et al. 2024). This attack method shows the transferability of adversarial examples between different models, which seriously threatens to the security and robustness of machine learning models.

### LLMs

LLMs have made remarkable progress and breakthroughs in recent years, and their capabilities have far exceeded those of traditional natural language processing (NLP) models. These models are trained by deep learning and large amounts of data to understand and generate natural language and perform various linguistic tasks such as text categorization, machine translation, Q&A, and text generation. Currently, LLMs represented by the GPT series (Brown et al. 2020; Radford et al. 2019; Ouyang et al. 2022; Achiam et al. 2023), ERNIE Bot (Sun et al. 2020), Flan-T5 series (Raffel et al. 2020; Chung et al. 2024), LLaMA (Touvron et al. 2023), GLM (Zeng et al. 2022), and UL2 (Tay et al. 2022) have been released globally. With the enrichment of LLMs corpora and the increase in parameter levels, LLMs have given rise to security risks such as value alignment (Porsdam Mann et al. 2023), model misuse (Picazo-Sanchez and Ortiz-Martin 2024),hallucination (Lin et al. 2022), prompt injection (Strobelt et al. 2022), and so on. In addition, as a deep learning model, LLMs are vulnerable to traditional attacks, such as adversarial attacks, data poisoning, and backdoor attacks. Researchers have experimentally demonstrated that LLMs are robust to adversarial attacks but are still affected by them, and that there is a certain degree of transferability of adversarial samples between LLMs (Wang et al. 2023); Shi et al. influenced the output of LLMs by using a backdoor attack to manipulate the reward model in the fine-tuning phase of RL (Shi et al. 2023). Unlike traditional NLP models, LLMs are accessed mainly through human-computer interaction through prompts. The elements of Prompt mainly include Instruction, Context, Input Data and Output Indicator. Instruction, as a necessary element, defines the task to be performed. The context is an optional element that provides information about the context or environment of the task; the input data is an optional element that provides data to be processed or analyzed; and the output indicator is another optional element that specifies the type or format of output.

CoT as emergent capability specific to LLMs, on the other hand, are stimulated by adding specific triggers or a step-by-step reasoning process to the prompt to stimulate the emergent capability of LLMs (Zoph et al. 2022); studies show that the larger the parameter level is, the more obvious the advantage of this capability is Wei et al. (2022). CoT is characterized by the transfer thinking of the same type of problems, which is usually used in combination with few-shot and zero-shot with ICL (Brown et al. 2020) to improve the model's inference performance. In existing research, Complexity-based prompting (Fu et al. 2022) enhances multi-step reasoning by using complex examples, which leads to improved accuracy for LLMs on complex tasks. Automatic CoT prompting (of Thought Prompting in Large Language Models 2023) reduces manual work through automation and adapts to various tasks, but the quality of the generated chains can be inconsistent. Synthetic prompting (Shao et al. 2023) enriches prompts by generating reasoning demonstrations and improving performance on complex tasks, but it relies on the quality of pre-training data, which may not fully align with task requirements. This paper adopts a semi-automated approach to construct the CoT by selecting samples from the same dataset and

utilizing LLMs to generate the reasoning process, thereby constructing the CoT. This method reduces manual effort while ensuring the CoT is aligned with the task requirements. It is worth noting that Li et al. (2023). adopted the idea of "Let's think step by step" from CoT and proposed a multi-step jailbreaking attack. Part of the attack successfully bypassed the security mechanisms in LLMs, leading to the generation of personal privacy information. They pointed out that the security of LLMs still needs improvement. While our work also focuses on exploring the security of LLMs, it differs from Li et al. (2023)'s approach. Their research emphasizes privacy attacks using CoT, but our study combines CoT with traditional adversarial perturbations to make the model reach incorrect conclusions. This combination helps us exploit model vulnerabilities in a new way.

In summary, our attack uses CoT techniques for adversarial sample attacks on LLMs. We apply traditional adversarial attack algorithms to a substitute model to generate perturbations, which are then transferred. Finally, we combine the CoT techniques with the transferred perturbations to create adversarial prompts. This decreases the large model's performance and accuracy. Through extensive experiments, this paper demonstrates our attack is more effective than previous approaches.

## The proposed attack

In this section, we introduce our proposed adversarial attack against LLMs. We begin by introducing our threat model and providing an overview of the attack framework. Then, we describe the various modules in detail.

### Threat models

In this paper, we assume that attackers operate under black-box conditions, where they only have access to the model's API and lack knowledge about its internal structure, parameters, or training process. Attackers first use a Transformer-based pre-trained model, fine-tuned for specific tasks, as a surrogate model. They apply existing adversarial attack algorithms to generate perturbation-capable samples. Then, we integrate the OpenAttack and PromptBench frameworks, utilizing CoT techniques and perturbation-capable adversarial samples. By employing various prompt engineering methods and perturbation strategies, we construct a CoT-based adversarial transfer attack framework to generate adversarial prompts. These adversarial prompts are stealthy,and when used to attack LLMs, they will cause the LLMs to produce wrong the output.

### CTTA framework

This framework implements textual adversarial attacks at word, character, and sentence levels through the OpenAttack attack framework and modifies the OpenAttack framework to obtain samples with perturbation capability. In addition, the CheckList and StressTest adversarial attack algorithms are extended in the OpenAttack attack framework based on the TextAttack (Morris et al. 2020) attack framework. Then, using the original data and the LLMs text generation interface, the task instructions are designed to generate the reasoning process for the initial task, which is the CoT of the LLMs on the realization of the initial task. Moreover, the transfer of adversarial samples on the CoT is realized by combining their specific task instructions according to different tasks. Finally, a black-box transfer-based attack is realized by using the LLM API interface. The overall framework is shown in Fig. 1.

### Adversarial attack module

This module mainly realizes textual adversarial attacks at different perturbation levels through the OpenAttack attack framework, to obtain adversarial samples with perturbation capability. OpenAttack is the first textual adversarial attack toolkit to support all attack types, covering all perturbation granularities and victim model Accessibility. Fully compatible with the Huggingface community's Transformer and Dataset libraries, supporting multi-language and multi-process attacks, and with high scalability and comprehensive review metrics, it is designed to assist researchers in rapid research and development related to textual adversarial attacks (Zeng et al. 2021). As shown in Table 1,the attack algorithms used in this module mainly includecharacter-level attacks: TextBugger, DeepWordBug; word-level attacks: BertAttack, TextFooler; and sentence-level attacks: CheckList, StressTest, the data are derived from the SST2 dataset, "1" is positive, "0" is negative.

The framework diagram of the module is shown in Fig. 2, the module's substitution models are fine-tuned language models on the input datasets, such as Bert, Roberta, etc. Different perturbation levels attacks satisfy the limits of the perturbation constraints during the attack, which have some differences depending on the algorithms, including semantic similarity, perturbation rate, word vector distance, cosine similarity, edit distance, and many other metrics, is used in the attack algorithm to evaluate, and adjust the generated adversarial samples. The module ultimately outputs adversarial samples that can spoof the substitution model, i.e., obtaining samples with perturbation capability.As shown in Eq. (1), the module utilizes the adversarial attack algorithm $\mathcal{L}$, attacks the substitute model s,and finds the worst perturbation $\delta$,the addition of perturbation to the clean sample $x$, so that the addition of perturbation to the clean sample $x$, leads to an incorrect output from the substitute model.

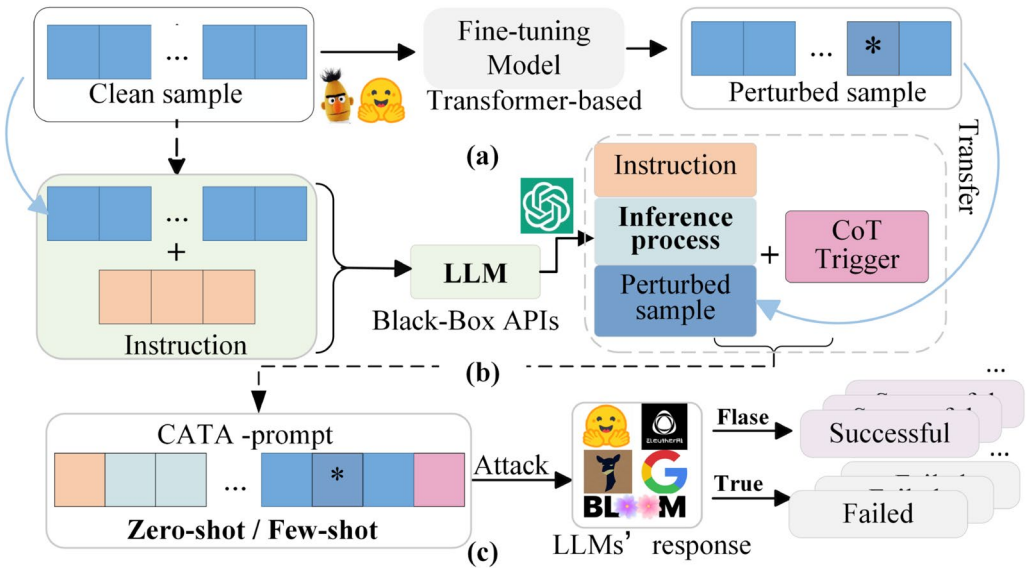


**Fig. 1** CTTA Framework Diagram, the framework consists of three modules that realize **a** Adversarial attack, **b** Transfer sample construction, and **c** Transfer-based attack

**Table 1** Examples of adversarial examples generated by different levels of attacks. Where the bold font is the attacked part, the underline font is the successful part of the attack

| Level | Attack | Samples | Label |
|---|---|---|---|
| character-level | TextBugger | A **compelling** motion picture that **illustrates** an American tragedy. | 1 |
| | | A compelnilg motion picture that illkstrates an American tragedy. | 0 |
| | DeepWordBug | It's a **day** at the **beach** – **with** air conditioning and **popcorn**. | 1 |
| | | It's a day at the beach –with air conditioning and popcorn. | 0 |
| word-level | BertAttack | **A** whole lot of **fun** and **funny** in the middle, **though** somewhat less hard - hitting at the start and finish. | 1 |
| | | a whole lot of long and moving in the middle, somewhat somewhat less hard - hitting at the start and finish. | 0 |
| | TextFooler | **Collateral Damage** is trash, but it **earns** extra **points** by **acting** as if it were n 't. | 0 |
| | | indirect price is trash, but it bring extra power by play as if it were n 't. | 1 |
| sentence-level | CheckList | Triple X is a double agent, and he's one bad dude. | 1 |
| | | Triple X is a double agent, and he's one bad dude vTAjHynolG. | 0 |
| | StressTest | You'd think by now America would have had enough of plucky British eccentrics with hearts of gold. | 0 |
| | | You'd think by now America would have had enough of plucky British eccentrics with hearts of gold and true is true. | 1 |

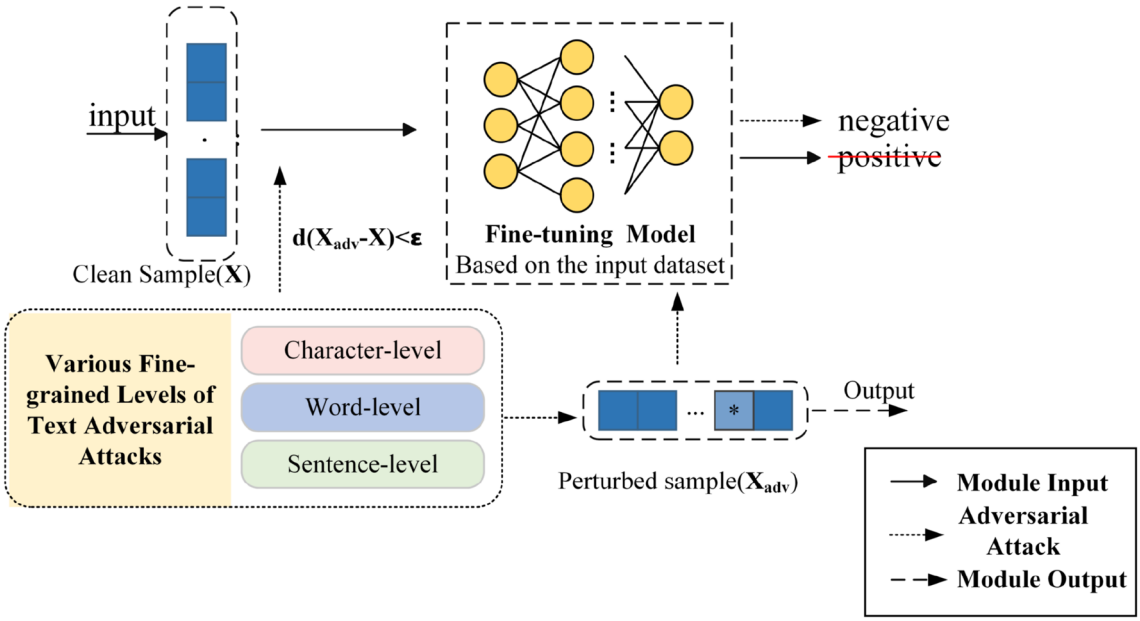


**Fig. 2** Adversarial attack module diagram, attack results take emotion classification task results as examples

$$\text{argmax}_{\delta \epsilon \varepsilon} \mathcal{L}[s(x+\delta), y]_{(x,y)\in\mathcal{Q}} \quad (1)$$

where $\mathcal{Q}$ is the original dataset, and $\varepsilon$ defines the perturbation range.

### Transfer sample construction module

The main function of this module is to realize the construction of adversarial transfer samples, using the adversarial samples with perturbations capability obtained in module a. Further, we construct adversarial migration prompt samples for LLMs based on CoT, and the framework diagram of this module is shown in Fig. 3. Among them, prompts are the mainstream way for users to interact with LLMs in a human-computer way, which mainly consists of four components: Instruction, Context, Input Data, and Output Indicator, where the instruction is the necessary information that determines the tasks that need to be performed by the LLMs, and the rest of them are optional elements.

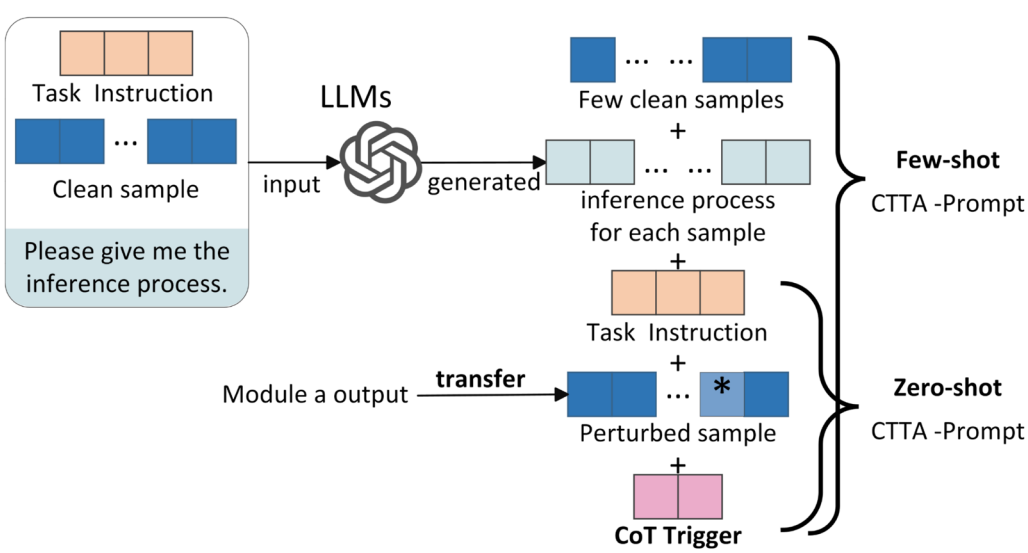


**Fig. 3** Transfer sample construction module diagram, construct few-shot and zero-shot CTTA-Prompt

To be specific,this module uses task data samples, task instructions, and specific output guidance as inputs for the LLM. It generates an inference process for the initial data samples based on correct conclusions. In this paper, GPT−3.5-Turbo is used as the LLM to implement this process. The generated reasoning process is added to few-shot as part of adversarial prompting. This provides reasoning samples to enhance the target model's CoT capability. The task instructions used in the prompt construction are derived from the optimal task prompt in PromptBench (Zhu et al. 2024), and different task types and prompt engineering types are matched with their respective optimal task instructions to ensure that the target LLMs can fully understand the task; the migration perturbation samples are derived from the output content in module a;and the CoT trigger is derived from the optimal trigger in the paper (Kojima et al. 2022). Assuming $I$ is the optimal task instruction and $f_1$ represents the GPT−3.5-Turbo large model, as shown in Eq. (2), the inference process is obtained.

$$\mathrm{Inp} = f_1(x + y + I + o) \tag{2}$$

where *Inp* represents the inference process, and o stands for the output instruction. As shown in Eqs.(3, 4), CTTA-prompt is derived.

$$\mathcal{A}(\mathrm{P}) = \mathrm{P} + \delta + \left[\sum \left(x_i, y_i, \mathrm{Inp}_i\right)\right]_{\gamma} \tag{3}$$

$$\mathrm{P} = \mathrm{I}_{(\mathrm{f},\mathcal{Q},\gamma)} + \mathrm{x} + \mathrm{CoT} \tag{4}$$

where $\mathcal{A}(\mathrm{P})$ denotes the CTTA-prompt, $\mathcal{A}$ is the adversarial attack applied to P, P stands for prompt, and $\gamma$ represents prompt engineering.

This paper utilizes two types of prompt engineering, few-shot and zero-shot, to construct CTTA-Prompt based on these two different prompt engineerings, respectively. Studies have shown (Wei et al. 2022; Radford et al. 2019; Schick and Schuetze 2021; Gao et al. 2021) that CoT prompts in both the few-shot and zero-shot modes are effective to vary degrees in solving the learning challenges in the case of limited or missing samples, and significantly improve the inference capability as well as the generalization performance of the model. Therefore, the use of these two constructs can maximize the potential of the CoT of LLMs, which in turn makes it easier to expose the shortcomings and deficiencies in the reasoning process of LLMs, thus revealing the flaws in the CoT capability of LLMs in a more direct way.

### Transfer-based attack module

The main function of this module is to implement transfer adversarial attacks against LLMs, by utilizing the adversarial prompts constructed in the module to trick the LLMs into obtaining erroneous outputs. The attack diagram for this module is shown in Fig. 4.

The module has both few-shot and zero-shot adversarial prompt samples, mainly through the API interface of LLMs for black-box attacks. If the model produces a different output from the clean sample, the attack is successful and results in an accuracy and performance drop of the model (Zhu et al. 2024). Among them, the final adversarial samples constructed in this paper are mainly perturbed against the input data portion of the prompt. The difference is that the PromptBench benchmark evaluation framework mainly perturbs the task instruction part of the prompt. Since optimal task instructions and CoT triggers are used in Module 2, model performance and accuracy are maximized on clean samples.Therefore, the objective of this module, as shown in Eq. (5), is to utilize the most effective adversarial attack, $\mathcal{A}$, from module b to induce erroneous outputs from the target large model.

$$\mathrm{argmax}_{\mathcal{A}}[f_\theta(\mathcal{A}, x), y]_{(\mathrm{x},\mathrm{y})\in\mathcal{Q}} \tag{5}$$

where $f_\theta$ is the LLM.

## Experiments and analysis

This section first introduces the target models and datasets, the experimental design and metrics, and the experimental results, and then concludes with analysis and discussion. The experiments are divided into two

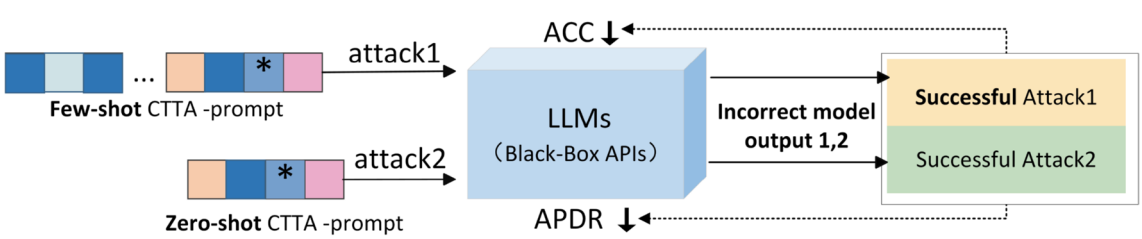


**Fig. 4** Transfer-based attack module diagram, indicating successful attack when the model produces incorrect results

parts: single-granularity perturbation experiments and mixed perturbation attack experiments.

In the single perturbation attack part, we focus on benchmark datasets such as SST-2, MNLI, and QNLI. We target LLMs like Flan-T5-large, UL2, and Vicuna, utilizing traditional adversarial attack algorithms such as TextBugger, DeepWordBug, BertAttack, TextFooler, CheckList, and StressTest to construct CTTA-prompt. These prompts enable character-level, word-level, and sentence-level transfer adversarial attacks. We thoroughly evaluate the changes in accuracy and performance metrics of the models before and after the attacks. Additionally, we compare the results of each fine-grained attack with baseline attacks to comprehensively verify the effectiveness and feasibility of our proposed attack strategies, while deeply analyzing their impact on model stability.

In the mixed perturbation attack part, we expand our study to more complex adversarial scenarios. We target models with different parameter scales, such as GPT-J-6B, GPT-NEOX-20B, BLOOM (176B), and ChatGPT (175B), using the adversarial AdvGLUE dataset. This dataset covers multiple tasks including SST-2, MNLI, QNLI, QQP, and RTE. We construct CTTA-prompt to implement mixed-granularity transfer adversarial attacks. The robustness of each model under attack is evaluated and compared to their robustness on the original AdvGLUE dataset. The goal is to demonstrate how this attack further weakens the adversarial robustness of LLMs, revealing new vulnerabilities in LLMs when facing adversarial threats.

### Target models and datasets

The target models of this paper involve a billion, 10 billion and 100 billion level model. Table 2 shows the target models involved in this paper and their information. The descriptions of the LLMs are as follows:

- Flan-T5-large is an extended version of the T5 (Text-to-Text Transfer Transformer) model developed by Google, focused on text generation tasks.
- Universal Language Model 2 (UL2) is a multi-task learning framework designed to enhance model generalization across various natural language processing tasks.
- Universal Language Model 2 (UL2) is a multi-task learning framework designed to enhance model generalization across various natural language processing tasks.
- GPT-J is a 6B parameter language model developed by EleutherAI, it excels in text generation and understanding tasks.
- GPT-NeoX-20B as a larger variant, demonstrates enhanced capability in complex text generation and processing tasks.
- BLOOM is a 176B parameter multilingual model developed by BigScience, designed to support language processing tasks across multiple languages.
- ChatGPT is a conversational model based on the GPT−3.5 architecture, developed by OpenAI. It is widely applied in various conversational tasks due to its strong generation and reasoning capability.

The statistics of the transfer samples used in each data set of this paper are presented in Tables 3 and 4.

Here, the total number of samples is equal to the sum of the number of transfer samples based on each attack multiplied by the number of prompt engineering types multiplied by the number of sample types multiplied by the number of LLMs.The descriptions of the relevant datasets are as follows:

- The SST-2 dataset consists of movie review sentences and the corresponding manually annotated sentiment. The goal is to predict the sentiment of a given review sentence, and the results are divided into positive and negative sentiments.

**Table 2** Statistics of the target LLMs in this paper, including their size, release organization, and release date

| Model | Size(B) | Release organization | Release date |
|---|---|---|---|
| Flan-T5-large (Raffel et al. 2020) | 0.78 | Google | Oct-2022 |
| UL2 (Tay et al. 2022) | 20 | Google | May-2022 |
| Vicuna (Zheng et al. 2024) | 13 | LM-SYS | Mar-2023 |
| GPT-J-6B (De la Rosaand and Fernández 2022) | 6 | EleutherAI | Jul-2021 |
| GPT-NEOX-20B (Black et al. 2022) | 20 | EleutherAI | Feb-2022 |
| BLOOM (Le Scao et al. 2023) | 176 | BigScience | May-2022 |
| ChatGPT (Brown et al. 2020) | 175 | OpenAI | Nov-2022 |

**Table 3** Statistics of the datasets with single perturbation level,and the number of transferred samples base on various attack algorithms

| Datasets | Test samples | character-level | | word-level | | sentence-level | | Total sample |
|---|---|---|---|---|---|---|---|---|
| | | TextBugger | DeepWordBug | BertAttack | TextFooler | CheckList | StressTest | |
| SST-2 | 1000 | 948 | 649 | 884 | 783 | 71 | 633 | 47616 |
| MNLI | 1000 | 597 | 241 | 93 | 236 | 53 | 44 | 15168 |
| QNLI | 1000 | 739 | 302 | 816 | 601 | 164 | 221 | 34116 |

Experimental datasets used source link https://gluebenchmark.com/tasks/

**Table 4** Statistics of datasets used by adversarial attacks with mixed perturbation levels

| Data sources | Datasets | Transfer samples | Total sample |
|---|---|---|---|
| AdvGLUE | SST-2 | 148 | 740 |
| | MNLI | 121 | 605 |
| | QNLI | 148 | 740 |
| | QQP | 78 | 390 |
| | RTE | 81 | 405 |

Experimental datasets used source link https://github.com/microsoft/robustlearn

- The MNLI dataset consists of sentence pairs for textual entailment. The task is to predict whether the premise sentence is related, contradictory,or neutral to the hypothesis sentence.
- The QNLI is derived from the classic reading comprehension task SQuAD question-answer dataset. The task is to predict whether a context sentence has an answer to a given question.
- The AdvGLUE is a dataset that integrates multiple tasks, such as sentiment classification, question answering, natural language inference, etc., and is generated by text attacks at different levels of granularity. In this paper, SST-2, MNLI, QNLI, QQP and RTE are selected. Among them, QQP is a repetitive sentence detection task dataset, and RTE is a textual entailment task dataset.

## Experimental design and metrics

In this paper, we first implement an transfer attack based on single perturbation level, Flan-T5-large, UL2, and Vicuna are used as target models,which uses DAC (drop accuracy), ADAC (average drop accuracy) and APDR (average performance drop rate) as the evaluation metrics of the experimental module.

Accuracy drop is the difference between the target model's accuracy on the clean sample and the adversarial sample, as shown in Eq. (6).

$$\mathrm{DAC}\left(\mathrm{P},\mathcal{A},f_\theta,\mathcal{Q}\right)=\left[\mathrm{Acc}\left(f_\theta(\mathrm{P},x),y\right)-\mathrm{Acc}\left(f_\theta(\mathcal{A}(\mathrm{P}),x),y\right)\right]_{(x,y)\in Q} \tag{6}$$

where $\mathcal{A}$ is the adversarial attack acting on P, P is the prompt sample, $f_\theta$ is a LLM, and $\mathcal{Q}$ is the data source.

The average accuracy drop is the average of the accuracy drop across all sets, as shown in Eq. (7).

$$\mathrm{ADAC}=\frac{1}{|\mathcal{M}|}\frac{1}{|\mathcal{P}|}\frac{1}{|\mathcal{Q}|}\mathrm{DAC}\left(P,\mathcal{A},f_\theta,\mathcal{Q}\right) \tag{7}$$

where $\mathcal{M}$ is the set of all models,$\mathcal{P}$ represents different types of prompt engineering, and $\mathcal{Q}$ is the set of all task types of datasets.

Performance drop rate (PDR) (Zhu et al. 2024) is a standardized metric to evaluate a large model under a hint adversarial attack, which quantifies the relative performance degradation after the attack, and the average performance drop rate (APDR) is the average performance degradation rate of a large model on different types of prompts and datasets, as shown in Eqs. (8) and (9).

$$\mathrm{PDR}\left(\mathcal{A},\mathrm{P},f_\theta,\mathcal{D}\right)=1-\frac{\sum_{(x;y)\in\mathcal{D}}\mathcal{M}[f_\theta([(\mathcal{A}(\mathrm{P}),x]),y]}{\sum_{(x;y)\in\mathcal{D}}\mathcal{M}[f_\theta([\mathrm{P},x]),y]} \tag{8}$$

$$\mathrm{APDR}=\frac{1}{|\mathcal{P}|}\frac{1}{|\mathcal{F}|}\frac{1}{|\mathbb{D}|}\sum_{\mathrm{P}\in\mathcal{P}}\sum_{f_\theta\in\mathcal{F}}\sum_{\mathcal{D}\in\mathbb{D}}\mathrm{PDR}\left(\mathcal{A},\mathrm{P},f_\theta,\mathcal{D}\right) \tag{9}$$

where $\mathcal{M}[\cdot]$is the evaluation function, which is equal to 1 if $\hat{y}=y$ and 0 otherwise. Note that a negative PDR means that adversarial prompts can occasionally boost the performance of LLMs. $\mathcal{P}$ is the set of all types of prompts, $\mathcal{F}$ is the set of all target models,$\mathbb{D}$ is the set of all datasets.

Secondly, in the experimental section on mixed-granularity attacks, this paper uses the Attack Success Rate (ASR) as the evaluation metric for adversarial robustness,as shown in Eq.(10).

$$\mathrm{ASR}=\sum_{(\mathrm{x},\mathrm{y})\in\mathcal{Q}}\frac{\mathbb{1}\mid f(\mathcal{A}(\mathrm{x})\neq\mathrm{y})\mid}{\mathbb{1}|f(\mathrm{x})=\mathrm{y}|} \tag{10}$$

where $\mathcal{Q}$ is the data source, $\mathcal{A}(x)$ represents the adversarial attack algorithm, $\mathbb{1}$ represents the indicator function, y represents the true label, and $f$ is the attack target model. Usually, ASR is used to evaluate the effectiveness of adversarial input.Notably, since AdvGLUE is constructed through multiple levels of adversarial attacks and manual creation, this part of the experiment does not focus on any single traditional adversarial attack algorithm.

As shown in Table 5, using word-level attacks as an example, the adversarial prompt samples designed in this experiment are presented, along with the differences from other prompt engineering methods,the adversarial prompt constructed by other perturbation level attacks is like this sample, with specific differences only in the disturbance part. In addition, the perturbation of PromptBench (Zhu et al. 2024) is only present in the instruction part of the prompt; the perturbation of Wang et al. (2023) exists in the part of the prompt data, and the parentheses are the part of the same task example under the few-shot prompt. The perturbation in this paper (CTTA) also exists in the data part, but the difference is that the task's inference process is added to the example part of the few-shot prompt constructed in this paper, and the trigger of the CoT is added to the prompt to stimulate the model's CoT ability. At the same time, the optimal task instruction is selected according to different task types and target models. That is, the task instruction can achieve the highest accuracy for the current task and model to minimize the impact of the task instruction on the performance of the model.

## Experimental results

Firstly, the results of the single perturbation fine-grained attack are presented. Tables 6, 7 and 8 show the ACC and DAC values of different LLMs, and Figs. 5, 6, and 7 show the APDR and ADAC values of LLMs under adversarial attacks of different perturbation levels.

As shown in Table 6, where CTTA values with advantage under the same attack, metric and datasets are bolded, Tables 7 and 8 is the same as Table 6.Compare with the results of PromptBench attack framework, the CTTA attack framework constructed in this paper has relatively good attack effects on the Flan-T5-Large model, the accuracy of most of the adversarial is significantly reduced, and the DAC value is significantly improved. As shown in Fig. 5,the change of PDR index of Flan-T5-Large under the CTTA attack framework is more stable than that of PromptBench attack mode, and under the perturbation data generated by TextBugger, DeepWordBug, BertAttack, TextFooler and StressTest adversarial attack algorithms, an APDR superior to PromptBench is obtained, under the perturbation data generated by TextBugger, BertAttack, TextFooler and CheckList adversarial attack algorithms, the ADAC is better than PromptBench.

As shown in Table 7,compared with the results of the PromptBench adversarial attack, the CTTA attack has better attack effect on the UL2 model, and the model is less adversarial and accurate, and the reduction amplitude is better than that of PromptBench attack. As shown in Fig. 6,under the adversarial prompt constructed by CTTA, the performance change is more stable than that of PromptBench, and the performance and accuracy of the model are significantly decreased. It even has a more obvious attack effect than the Flan-T5-Large model with smaller parameter level.

As shown in Table 8,although the accuracy rate of Vicuna model on CTTA adversarial prompt samples is obviously lower than that of the previous two LLMs and

**Table 5** Samples of adversarial prompts constructed by different prompt engineering techniques. Where the bold font is the perturbation caused by adversarial attacks with different perturbation levels, and the underline font is the new part of this paper

| Prompt engineering | Sample of adversarial prompts |
|---|---|
| Clean | Are the following two questions equivalent or not? Answer me with "equivalent" or "not _ equivalent".Q1..., Q2... |
| Zhu et al. (2024) | {Here are three examples...}. Are the **so a** equations equivalent or not? Answer me with "equivalent" or "not_equivalent". Q1..., Q2... |
| Wang et al. (2023) | Are the following two questions equivalent or not? Answer me with "equivalent" or "not_equivalent". Q1: What are the best things to do in **Hog** Kong? Q2: What is the best thing in Hong Kong? |
| CTTA | {Here are three examples...+inference process}. Are the following two questions equivalent or not? Answer me with "equivalent" or "not_equivalent". Q1: What are the best things to do in **Hog** Kong? Q2: What is the best thing in Hong Kong?Let's think step by step. |

**Table 6** Adversarial attack results of Flan-T5-Large model, where CTTA values with advantage under the same attack, metric, and datasets are bolded

| Dataset | Attack | | TextBugger | | DeepWordBug | | BertAttack | | TextFooler | | CheckList | | StressTest | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | metrics(%) | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA |
| SST | ZS | ACC↓ | 95.41 | **71.81** | 94.04 | **77.66** | 89.79 | **58.26** | 38.07 | 68.45 | 94.72 | **69.01** | 94.72 | **82.94** |
| | | DAC↑ | 1.26 | **22.34** | 1.38 | **13.87** | 5.62 | **33.48** | 57.34 | 23.12 | 0.69 | **7.04** | 0.69 | **7.42** |
| | FS | ACC↓ | 94.95 | **71.28** | 94.95 | **78.89** | 94.04 | **60.40** | 94.27 | **69.86** | 95.30 | **67.61** | 95.41 | **85.47** |
| | | DAC↑ | 0.57 | **22.34** | 0.57 | **12.02** | 1.49 | **30.54** | 1.26 | **20.95** | 0.23 | **7.04** | 0.11 | **4.11** |
| MNLI | ZS | ACC↓ | 81.90 | **41.37** | 94.04 | **77.66** | 81.40 | **27.96** | 82.40 | **38.56** | 85.50 | **52.83** | 68.30 | **50.00** |
| | | DAC↑ | 4.10 | **39.87** | 1.38 | **13.87** | 4.60 | **53.76** | 3.60 | **44.07** | 0.50 | **16.98** | 17.70 | 15.91 |
| | FS | ACC↓ | 85.60 | **40.37** | 94.95 | **51.45** | 62.50 | **27.96** | 83.10 | **37.29** | 86.20 | **54.72** | 63.90 | **59.09** |
| | | DAC↑ | 1.60 | **41.04** | 0.57 | **24.07** | 24.70 | **58.06** | 4.10 | **44.49** | 1.00 | **13.21** | 23.30 | **4.55** |
| QNLI | ZS | ACC↓ | 56.80 | 84.98 | 38.60 | 91.39 | 42.20 | 77.94 | 41.40 | 82.70 | 89.00 | **88.41** | 88.70 | **82.94** |
| | | DAC↑ | 33.00 | 5.68 | 51.20 | 1.66 | 47.60 | 12.75 | 48.40 | 9.48 | 0.80 | **3.66** | 1.10 | **7.42** |
| | FS | ACC↓ | 78.80 | 79.97 | 39.50 | 91.06 | 45.50 | 73.16 | 82.70 | 78.04 | 88.50 | 89.63 | 84.20 | 85.07 |
| | | DAC↑ | 10.50 | **10.69** | 49.80 | 1.99 | 43.80 | 16.42 | 6.60 | **13.64** | 0.80 | **3.66** | 5.10 | **7.24** |

Here "PromptBench" Zhu et al. (2024) is simply "Bench"

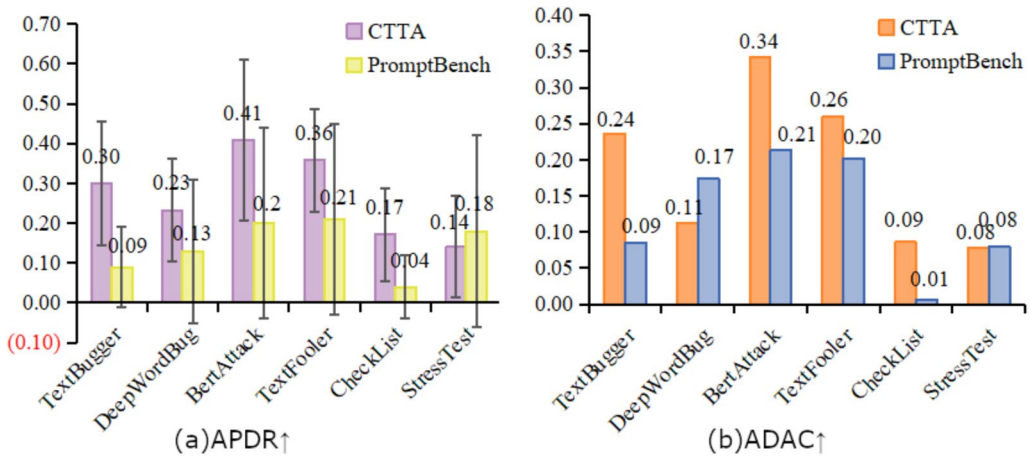


**Fig. 5** APDR(**a**) and ADAC(**b**) values of the Flan-T5-Large model under different attack levels

has a certain amplitude of accuracy decline, it is still inferior to the adversarial prompt samples constructed in PromptBench on the whole; As shown in Fig. 7, for the Vicuna model, the average performance drop rate and average accuracy drop of the model caused by different types of disturbances by the CTTA attack framework are lower than those of the PromptBench attack framework. This special performance is analyzed and discussed in the next subsection.

Secondly, in the mixed fine-grained attack experiment,Fig. 8 shows the attack success rates of four LLMs -GPT-J-6B, GPT-NEOX-20B, BLOOM, and ChatGPT (175B)-using CTTA adversarial prompts.

The results indicate that the CTTA method achieves higher attack success rates across all models and tasks compared to Wang et al.'s findings. This demonstrates that CTTA can effectively weaken the adversarial robustness of LLMs. For the GPT-J-6B model, CTTA significantly increased the ASR across all tasks, especially exceeding 90% in the MNLI and RTE tasks. GPT-NEOX-20B showed noticeable improvements with CTTA in the QQP and MNLI tasks, though gains are more limited in other tasks. BLOOM had the most significant improvement in QQP, with ASR close to 100%, and showed substantial gains in other tasks. Although ChatGPT displayed some robustness against adversarial attacks with relatively low ASR, the CTTA method still significantly increased the attack success rate, especially in the SST2 and RTE tasks.

**Table 7** Adversarial attack results of UL2 model, where CTTA values with advantage under the same attack, metric, and datasets are bolded

| Dataset | Attack | | TextBugger | | DeepWordBug | | BertAttack | | TextFooler | | CheckList | | StressTest | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | metrics(%) | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA |
| **SST** | ZS | ACC↓ | 95.87 | **82.91** | 95.87 | **77.66** | 95.41 | **58.37** | 95.53 | **68.58** | 95.99 | **69.01** | 73.85 | 82.94 |
| | | DAC↑ | 0.69 | **10.34** | 0.69 | **13.87** | 1.15 | **33.37** | 1.03 | **22.99** | 0.57 | **7.04** | 22.71 | 7.42 |
| | FS | ACC↓ | 96.10 | **79.11** | 96.22 | **81.66** | 95.64 | **63.69** | 96.10 | **72.67** | 96.56 | **69.01** | 96.67 | **84.68** |
| | | DAC↑ | 0.80 | **13.08** | 0.69 | **8.94** | 1.03 | **27.60** | 0.69 | **18.13** | 0.34 | **7.04** | 0.23 | **9.64** |
| **MNLI** | ZS | ACC↓ | 86.50 | **48.74** | 87.40 | **58.92** | 73.70 | **26.88** | 81.30 | **35.59** | 88.30 | **58.49** | 89.50 | **52.27** |
| | | DAC↑ | 2.30 | **26.80** | 1.40 | **14.52** | 15.10 | **53.76** | 7.50 | **45.34** | 0.50 | **7.55** | -0.70 | **13.64** |
| | FS | ACC↓ | 88.10 | **61.14** | 88.30 | **68.88** | 82.70 | **27.96** | 85.60 | **47.46** | 89.00 | **58.49** | 89.40 | **50.00** |
| | | DAC↑ | 1.00 | **24.79** | 0.90 | **8.71** | 6.30 | **56.99** | 3.50 | **36.44** | 0.20 | **7.55** | -0.20 | **13.64** |
| **QNLI** | ZS | ACC↓ | 93.40 | **87.69** | 94.20 | **90.40** | 89.30 | **79.66** | 93.40 | **84.53** | 94.50 | **89.63** | 94.20 | **90.50** |
| | | DAC↑ | 1.20 | **3.79** | 0.40 | **2.32** | 5.30 | **12.13** | 1.20 | **7.15** | 0.10 | **1.83** | 0.40 | **2.26** |
| | FS | ACC↓ | 93.80 | **88.77** | 93.90 | **89.40** | 92.20 | **80.27** | 93.30 | **83.03** | 94.20 | **90.85** | 94.60 | **87.78** |
| | | DAC↑ | 0.70 | **3.38** | 0.60 | **3.31** | 2.30 | **12.50** | 1.30 | **9.98** | 0.30 | **0.61** | -0.10 | **4.52** |

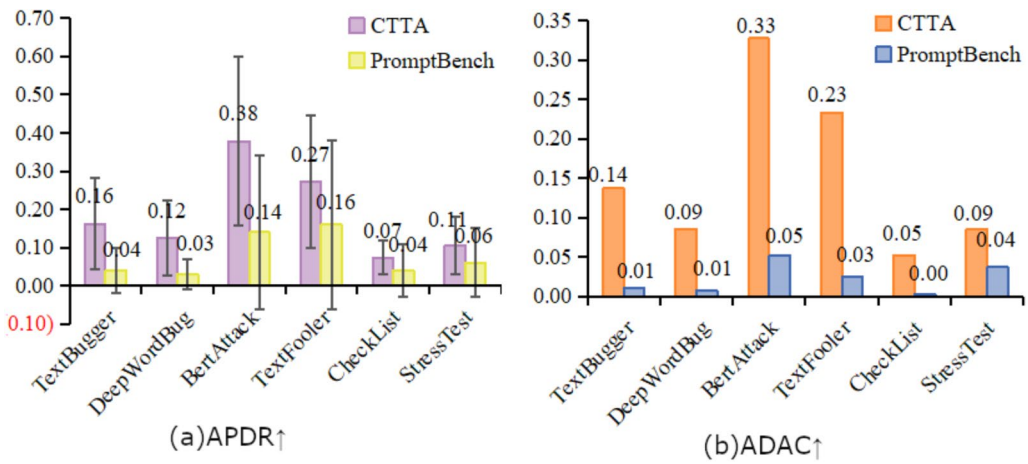


**Fig. 6** APDR(**a**) and ADAC(**b**) values of the UL2 model under different attack levels

**Table 8** Adversarial attack results of Vicuna model

| Dataset | Attack | | TextBugger | | DeepWordBug | | BertAttack | | TextFooler | | CheckList | | StressTest | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | metrics(%) | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA | Bench | CTTA |
| SST | ZS | ACC↓ | 11.12 | 49.16 | 3.10 | 38.06 | 5.85 | 25.79 | 3.21 | 26.82 | 0.00 | 30.99 | 17.89 | 44.23 |
| | | DAC↑ | 42.89 | 10.13 | 50.92 | 16.95 | 48.05 | 31.00 | 50.69 | 28.74 | 53.90 | 7.04 | 36.01 | 13.42 |
| | FS | ACC↓ | 0.57 | 32.38 | 0.57 | 16.80 | 0.11 | 11.87 | 2.87 | 12.13 | 4.47 | 19.72 | 3.78 | 17.22 |
| | | DAC↑ | 32.80 | 11.71 | 32.80 | 21.42 | 33.26 | 29.07 | 30.50 | 31.29 | 28.90 | 7.04 | 29.59 | 24.33 |
| MNLI | ZS | ACC↓ | 1.80 | 41.37 | 13.90 | 40.66 | 3.30 | 40.86 | 15.00 | 39.83 | 15.00 | 39.62 | 17.89 | 44.23 |
| | | DAC↑ | 27.40 | 7.37 | 15.40 | 6.22 | 25.90 | 1.08 | 14.20 | 10.59 | 14.20 | 7.55 | 36.01 | 13.43 |
| | FS | ACC↓ | 0.30 | 23.28 | 4.30 | 24.48 | 0.90 | 23.66 | 1.10 | 25.00 | 3.70 | 26.42 | 3.78 | 17.22 |
| | | DAC↑ | 24.80 | 13.90 | 20.80 | 13.69 | 24.20 | 15.05 | 24.00 | 13.98 | 21.40 | 1.89 | 29.59 | 24.33 |
| QNLI | ZS | ACC↓ | 0.10 | 42.90 | 6.20 | 35.76 | 0.40 | 39.88 | 1.20 | 41.93 | 10.80 | 37.20 | 17.89 | 44.23 |
| | | DAC↑ | 38.30 | 14.88 | 32.30 | 25.83 | 38.00 | 18.94 | 37.20 | 17.47 | 27.60 | 21.95 | 36.01 | 13.43 |
| | FS | ACC↓ | 1.20 | 45.06 | 1.80 | 42.72 | 0.30 | 44.36 | 0.20 | 47.59 | 3.70 | 44.51 | 3.78 | 17.22 |
| | | DAC↑ | 41.70 | 2.84 | 41.10 | 2.32 | 42.60 | 4.90 | 42.70 | 2.16 | 39.20 | 7.32 | 29.59 | 24.33 |

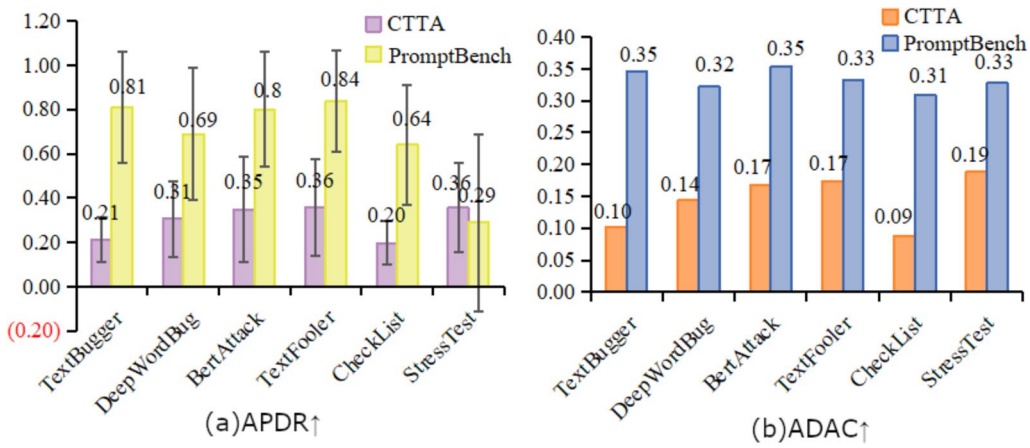


**Fig. 7** APDR(**a**) and ADAC(**b**) values of the Vicuna model under different attack levels

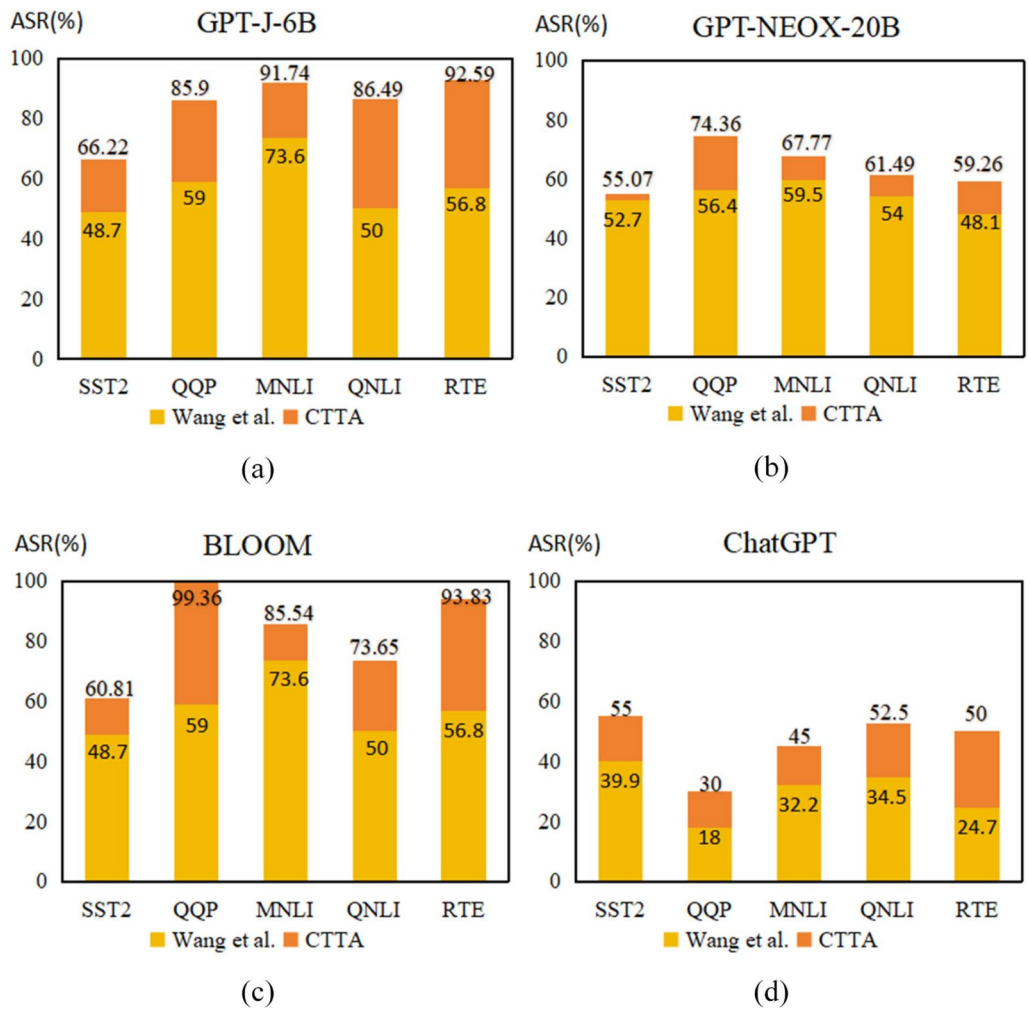


**Fig. 8** Adversarial robustness↓ of LLMs of different sizes based on different adversarial attacks

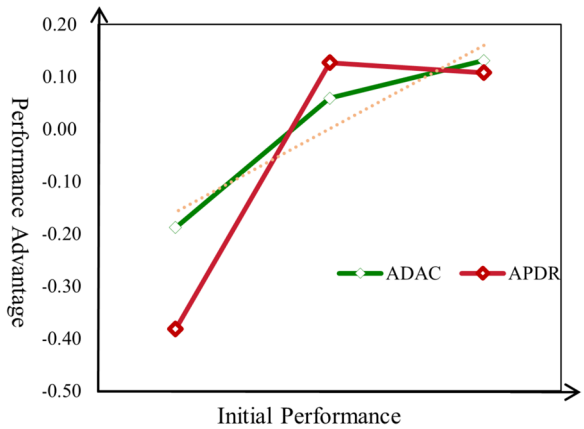


**Fig. 9** Advantages of CTTA attack with different metrics under different initial performance

## Analysis and discussions

Experimental results demonstrate that the CTTA-Prompt leads to a decline in performance and accuracy in the Flan-T5-large and UL2 LLMs across various perturbation granularities, outperforming existing adversarial prompt techniques. Notably, while the Vicuna model experiences some reduction in accuracy and performance under the CTTA-Prompt attack, the effect is not as pronounced as that observed in Flan-T5-large and UL2. To analyze this phenomenon, we calculated the initial performance of the models and the attack advantage of CTTA on these models, as illustrated in Fig. 9, the initial performance is positively correlated with the attack advantage. As noted in existing research (He et al. 2021) models with higher capability are more vulnerable to adversarial transfer attacks, whereas models with lower initial performance are less affected. As shown in Fig. 10a, the initial performance of the Vicuna model is significantly lower than that of Flan-T5-large and UL2, which explains the lower APDR and ACDC metrics for Vicuna, aligning with existing research findings. Additionally, as depicted in Fig. 10b,the adversarial robustness of vicuna against CTTA-Prompt is lower compared to guessing (Wang et al. 2023) results, indicating that our attack remains effective on the Vicuna model. Therefore, our attack method is valid and feasible, with models exhibiting better initial performance being more susceptible to the attack.The injection of chain-of-thought reasoning leads to a further decrease in the adversarial robustness of LLMs, increasing their vulnerability to adversarial attacks. This is especially evident in advanced models with high robustness, such as ChatGPT, where adversarial robustness on specific tasks decreases significantly. This also indicates that while LLMs continue to improve their performance, they may become more sensitive to perturbations. Moreover, the activation of chain-of-thought capability further exposes this vulnerability.

### *Sample quality evaluation*

Existing research points out that the definition of text perturbation $d()$ must consider semantic similarity and syntactic fluency (Wang et al. 2022) to ensure the quality of adversarial samples. Therefore, this study uses the following evaluation indicators to evaluate the sample quality of CTTA-Prompt. Cosine Similarity (CosSim) is used to assess the semantic similarity between adversarial samples and clean samples, and the sentence fluency of Perplexity (PPL) measurement calculated by GPT-2 was used, Grammar Error Rate (GErr) is calculated using the grammar errors provided by the Language Tool. In this paper, 360 samples are randomly selected to calculate each measurement index for each LLM.The number of samples is equal to the number of prompt engineering times the number of perturbation algorithms times the number of randomly drawn samples times the number of cycles,that's 2*6*10*3=360.

As shown in Fig. 11a, the cosine similarity between the CTTA adversarial samples and the clean samples are all above 0.85, which is usually between 0.7 and 0.85 in the classical adversarial attack algorithm (Li et al. 2020; Zang et al. 2020; Garg and Ramakrishnan 2020). In advanced adversarial attack algorithms (Ni et al. 2024), this index typically fluctuates around 0.85. The AdvGLUE dataset used in the adversarial experiments of Bloom, GPT-NeoX, and GPT-J LLMs is added with the constraint of similarity in the construction process to ensure the semantic similarity of the adversarial samples. Therefore, the adversarial examples constructed in this paper preserve the semantics of the clean examples.

As shown in Fig. 11b, the PPL value of CTTA adversarial samples on zero-shot is between 120 and 180, and the PPL value on few-shot is between 20 and 40. Currently, the PPL value of the adversarial samples generated by the advanced adversarial attack algorithm is between 60 and 230 (Ni et al. 2024; Liu et al. 2022). The PPL value of the adversarial samples generated by the classical adversarial sample attack algorithm is between 90 and 350 (Li et al. 2020; Zang et al. 2020; Garg and Ramakrishnan 2020), which proves that the adversarial samples generated in this paper maintain the fluency of the sentences.

As shown in Fig. 11c, the grammar error rate of the adversarial samples constructed in this paper is less than 0.12, the GErr value of the adversarial samples generated by the most advanced adversarial attack algorithm is between 0.10 and 0.20 (Ni et al. 2024), and the classical adversarial example attack algorithm is between 0.10 and 0.35 (Li et al. 2020; Zang et al. 2020; Garg and

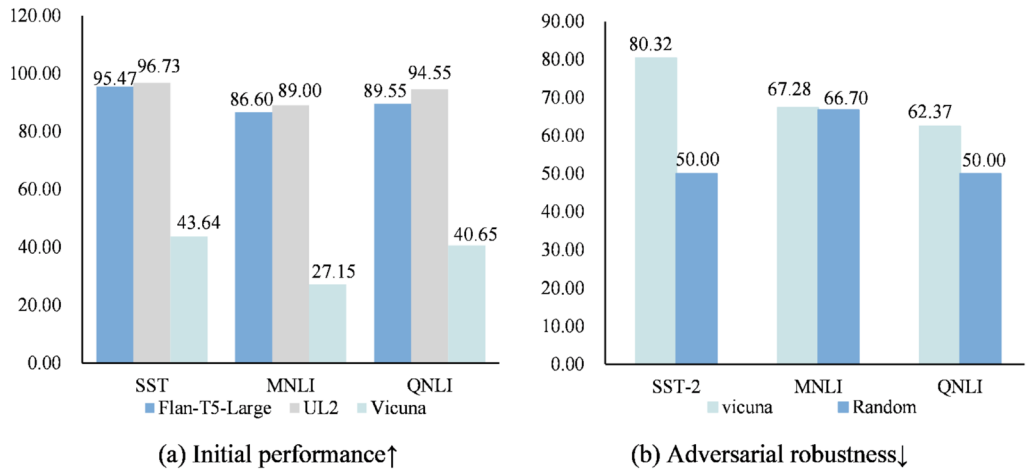


**Fig. 10** Comparison of Vicuna model performance

Ramakrishnan 2020). This indicates that CTTA-Prompt has grammatical accuracy.

In short, compared with the quality of the adversarial examples generated by the current advanced and classical adversarial attack algorithms, the quality of the CTTA-Prompt has semantic similarity and syntactic fluency, that is, the adversarial example is imperceptible.

### *Attention visualization*

To deeply explore the attention changes caused by samples that can successfully attack the model inside the model, this paper uses the gradient attention technology to obtain the attention scores on LLM (Zhu et al. 2024). Specifically, this technique assigns an attention score to each word based on the gradient, thereby revealing the dynamic changes in the model's attention when processing these samples. As shown in Fig. 12, the attention distribution of the original and adversarial samples on LLM with different perturbation leves is demonstrated. The darker the color is, the higher the attention weight is and the more significant influence on the response results of the model. Among them T is the correct model response and F is the incorrect model response, we used the Flan-T5-Large model here. In character-level attacks, it can be verified through the attention distribution of the original samples that the vital information affecting the results lies in the generated content constraint part in the instructions and the words such as "level", "lose", "them", "next", "you" and "recall" in the data. When perturbation is added to words such as "level", "lose", "them" and "recall" in the adversarial sample, the attention distribution is significantly changed on "lose" and "them" with large perturbation, which changes the key information extracted by the large model on the text, resulting in erroneous output. In the word-level attack, the word "Data" in the original sample is changed into "graph" through the word-level attack, and the original

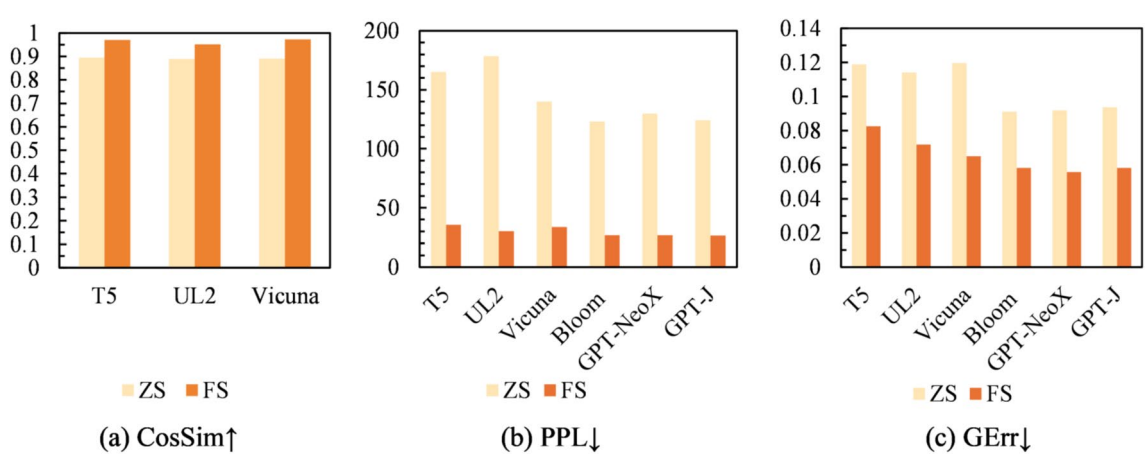


**Fig. 11** Indexes of adversarial samples for different LLMs

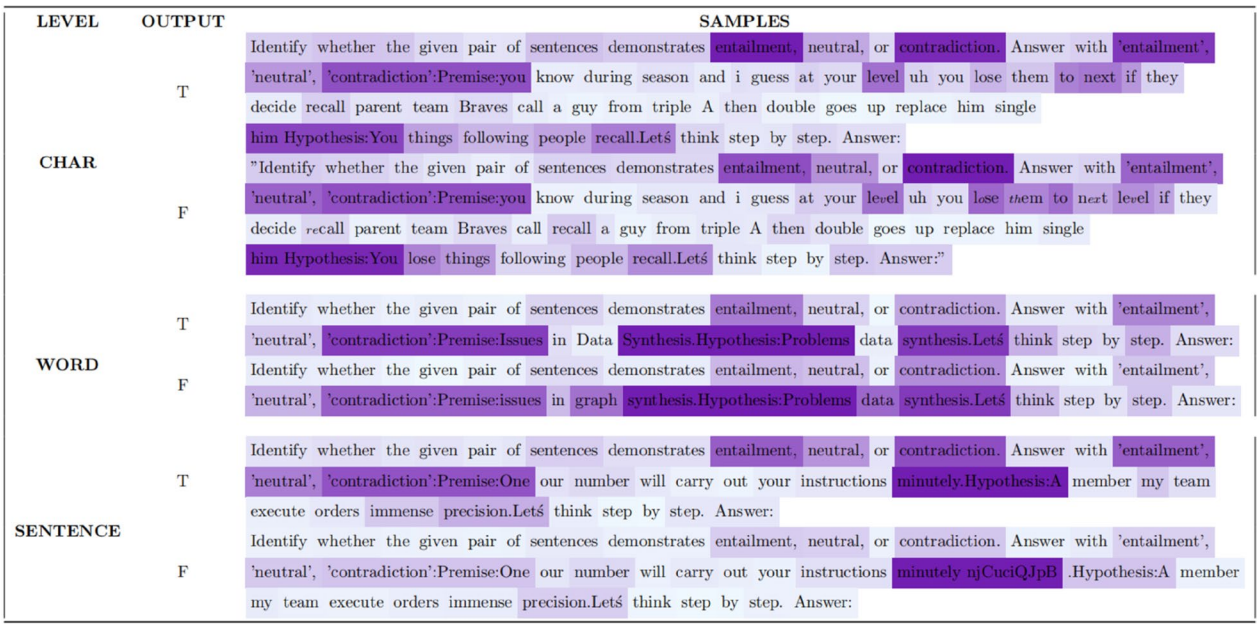


**Fig. 12** Attention distribution in the CTTA construct samples in sentence-level attacks

attention weight is also significantly improved.At the same time, the attention weight of "data" in "Hypothesis" is also considerably increased, which indicates that the change of attention weight caused by adding disturbance is global. In the sentence-level perturbation process, attention changes are more obvious, and the irrelevant string added at the end of the "Premise" sentence becomes the main information affecting the LLM response. In addition, as shown in Fig. 13, taking sentence-level attacks as an example, this paper compares the attention changes before and after the adversarial samples in PromptBench, and compares them with the adversarial samples constructed in this paper to analyze the role of CoT in the model response. The perturbed part of the sample obtains a certain degree of attention distribution, but the attention distribution of the rest of the words does not change significantly. Compared with the original sample constructed in this paper, the attention distribution of the sample in LLM not only lies in the attention to the important words in the data part, but also the attention distribution of the keywords in the instruction part is significantly higher than the corresponding part in PromptBench, which indicates that the increase of the CoT trigger makes the attention distribution of LLM more reasonable and the generated content more accurate when processing the text. Compared with the adversarial samples constructed in this paper, the attention distribution of the adversarial samples constructed in this paper is focused on the added perturbation part, which is significantly different from the attention distribution in PromptBench, indicating that the CoT capability makes LLM more sensitive to the perturbation and more accessible to be misled.

## Defenses

In this section, we will discuss potential defense mechanisms against CTTA. We will approach the discussion from three stages of LLMs: the fine-tuning phase, the input preprocessing phase, and the output phase.

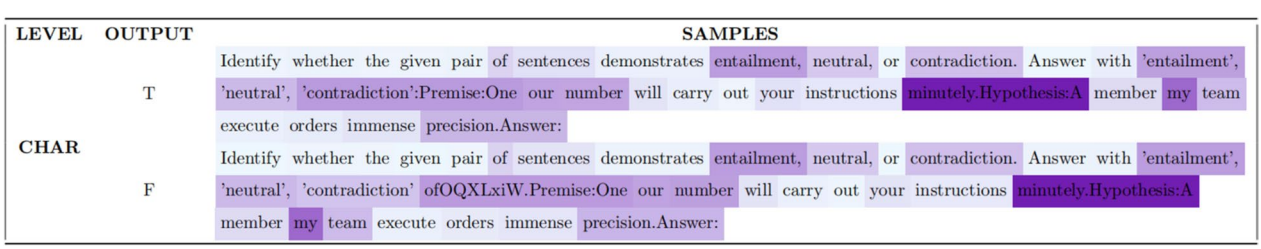


**Fig. 13** Attention distribution in the PromptBench construct samples in sentence-level attacks

- During the fine-tuning phase of LLMs, we propose incorporating various types of adversarial inputs into the training dataset. This approach enables LLMs to learn from a diverse array of adversarial examples. By doing so, the model can develop a more robust understanding of different adversarial patterns and tactics. Specifically, the introduction of such inputs can enhance the LLM's ability to autonomously detect and identify adversarial text embedded within otherwise normal contexts. This capability is particularly crucial for mitigating CTTA, as these attacks often disguise adversarial content within seemingly plausible contexts. Furthermore, regular updates to the adversarial input data should be conducted to ensure that LLMs can adapt to new and evolving attack strategies.
- At the input preprocessing stage, we recommend employing techniques such as spell-checking, syntactic analysis, and perplexity evaluation to thoroughly examine each input sentence. Spell-checking can help identify subtle manipulations in word forms that may be introduced during an adversarial attack, while perplexity evaluation assesses the overall likelihood or fluency of the input. Sentences with unusually high perplexity scores or atypical syntactic structures should be flagged for further review. This additional layer of examination can involve passing the flagged input through a secondary fine-tuned model specifically trained to recognize adversarial patterns. By detecting abnormal or suspicious inputs at this early stage, we can effectively reduce the risk of adversarial attacks before they reach the LLM's core processing pipeline.
- In the output phase, we propose requiring LLMs to generate outputs accompanied by a detailed reasoning process, following techniques such as Chain-of-Thought (CoT) prompting. This approach encourages the LLM to break down its reasoning step by step, making the thought process behind the final output more transparent and easier to review. Additionally, outputs with high attention weights, which indicate critical portions of the input influencing the output, can be highlighted. These flagged outputs should undergo either manual or automated review, particularly in high-risk scenarios. By examining the reasoning and attention weights, it becomes easier to identify instances where adversarial manipulation may have influenced the LLM's decision-making, thereby allowing for appropriate mitigation strategies.

## Conclusion

This paper presents a framework for adversarial example transfer attacks based on CoT for LLMs and investigates the performance of LLMs on adversarial prompts. The experimental results demonstrate the superiority of the attack framework compared to current advanced research, the effectiveness of CTTA, and its universal applicability to LLMs of different sizes. Through analysis, the following conclusions are drawn: the initial performance of LLMs is directly proportional to the degree of performance degradation under attack because CoT triggers more easily stimulate LLMs with higher initial performance and are more sensitive to sample perturbation.For LLMs with poor performance, their behavior aligns with existing research findings. In other words, the sensitivity of transfer adversarial samples is low, and the advantage of CTTA attack is not apparent. However, it still maintains a certain level of attack success rate. Additionally, the stimulation of CoT capability leads to a decrease in the model's adversarial robustness. The experimental results indicate that this influence is related to the scale of LLM parameters and the model's performance and CoT capability.Furthermore, this paper uses key indicators to compare the quality of adversarial examples generated by current advanced and classic adversarial example attack algorithms. This comparison demonstrates that constructed examples' semantic similarity and syntactic fluency ensure sample quality. Moreover, gradient attention is utilized in this paper to analyze the error response of LLMs to adversarial prompts deeply. It was found that through stimulation of CoT capability, LLMs exhibit more pronounced differences in attention distribution on prompts - specifically showing more explicit keyword extraction during prompt comprehension and reasoning processes. While this may result in higher performance indices for original samples, it also renders LLM attention distribution more sensitive to perturbed parts within samples - making them more susceptible to adversarial attacks.

In conclusion, LLMs' security and development complement each other. although the CoT capability of LLMs can enhance their robustness to some extent, they are also vulnerable to CoT-based adversarial attacks. This susceptibility leads to a rapid decline in model performance, reduced accuracy, and the generation of erroneous outputs, thereby exposing the weakness of LLMs in their CoT capability.So,The attack study in this paper helps to improve the LLMs' security mechanisms.In addition,due to the limitations of resources and time, it is difficult to evaluate all LLMs, which affects the depth and breadth of research to a certain extent. The target model of this study is mainly the general LLMs and fails to use the domain task-specific

data set to study the domain LLMs to discuss the general applicability of the attack further. Since the CoT capability of LLMs makes our attack more effective, we believe that in complex and dynamic tasks, CoT construction methods like Complexity-based prompting, Auto-CoT, and Synthetic prompting could lead to better attack results, which will be further investigated in our future work.

**Abbreviations**

| | |
|---|---|
| LLMs | Large language models |
| CoT | Chain-of-thought |
| CTTA | CoT Transfer adversarial attack framework |
| ICL | In-context learning |
| DAC | Drop accuracy |
| ADAC | Average Drop accuracy |
| PDR | Performance Drop rate |
| APDR | Average Performance drop rate |
| ASR | Attack Success rate |

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**Availability of data and materials**
Publicly available data used in this study was obtained from https://glueebenchmark.com/tasks/, https://github.com/microsoft/robustlearn, https://adversarialglue.github.io/, https://github.com/microsoft/promptbench, and case data will be made available upon request.

## Declarations

**Ethics approval and consent to participate**
Not applicable.

**Competing interests**
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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**References**

Shanahan M (2024) Talking about large language models. Commun ACM 67(2):68–79. https://doi.org/10.1145/3624724
Vaswani A, Shazeer N, Parmar N, et al (2017) Attention is all you need. In: Proceedings of the 31st Annual Conference on Neural Information Processing Systems ,NeurIPS, p 6000-6010, https://dl.acm.org/doi/10.5555/3295222.3295349
Brown T, Mann B, Ryder N et al (2020) Language models are few-shot learners. Adv Neural Inf Process Syst 33:1877–1901 (https://dl.acm.org/doi/abs/10.5555/3495724.3495883)
Chang Y, Wang X, Wang J et al (2024) A survey on evaluation of large language models. ACM Trans Intell Syst Technol 15(3):1–45. https://doi.org/10.1145/3641289
Zoph B, Raffel C, Schuurmans D, et al (2022) Emergent abilities of large language models. Transactions on Machine Learning Research https://openreview.net/forum?id=yzkSU5zdwD
Wei J, Wang X, Schuurmans D et al (2022) Chain-of-thought prompting elicits reasoning in large language models. Adv Neural Inf Process Syst 35:24824–24837 (https://dl.acm.org/doi/10.5555/3600270.3602070)
Li Z, Wang W, Li J et al (2024) Ucg: a universal cross-domain generator for transferable adversarial examples. IEEE Trans Inf Forensics Secur 19:3023–3037. https://doi.org/10.1109/TIFS.2024.3352913
Carlini N, Wagner D (2017) Towards evaluating the robustness of neural networks. In: Proceedings of the 2017 ieee symposium on security and privacy (sp), Ieee, pp 39–57, https://doi.org/10.1109/SP.2017.49
Moosavi-Dezfooli SM, Fawzi A, Fawzi O, et al (2017) Universal adversarial perturbations. In: Proceedings of the IEEE conference on computer vision and pattern recognition,CVPR, pp 1765–1773, https://doi.org/10.1109/CVPR.2017.17
Zhang J, Qian W, Nie R et al (2023) Generate adversarial examples by adaptive moment iterative fast gradient sign method. Appl Intell 53(1):1101–1114. https://doi.org/10.1007/s10489-022-03437-z
Zhao Z, Dua D, Singh S (2018) Generating natural adversarial examples. In: Proceedings of the 6th International Conference on Learning Representations,ICLR, https://iclr.cc/virtual/2024/20965
Gao J, Lanchantin J, Soffa ML, et al (2018) Black-box generation of adversarial text sequences to evade deep learning classifiers. In: Proceedings of the 2018 IEEE Security and Privacy Workshops,SPW, IEEE, pp 50–56, https://doi.org/10.1109/SPW.2018.00016
Li J, Ji S, Du T, et al (2019) Textbugger: Generating adversarial text against real-world applications. In: Proceedings of the 26th Annual Network and Distributed System Security Symposium,NDSS, https://dx.doi.org/10.14722/ndss.2019.23138
Li L, Ma R, Guo Q, et al (2020) Bert-attack: Adversarial attack against bert using bert. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing ,EMNLP, pp 6193–6202, https://doi.org/10.18653/v1/2020.emnlp-main.500
Jin D, Jin Z, Zhou JT, et al (2020) Is bert really robust? a strong baseline for natural language attack on text classification and entailment. In: Proceedings of the AAAI conference on artificial intelligence, pp 8018–8025, https://doi.org/10.1609/aaai.v34i05.6311
Kenton JDMWC, Toutanova LK (2019) Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT, p 4171-4186, https://doi.org/10.18653/v1/N19-1423
Naik A, Ravichander A, Sadeh N, et al (2018) Stress test evaluation for natural language inference. In: Proceedings of the 27th International Conference on Computational Linguistics,ACL, pp 2340–2353, https://aclanthology.org/C18-1198
Bhatt S, Jain R, Dandapat S, et al (2021) A case study of efficacy and challenges in practical human-in-loop evaluation of nlp systems using checklist. In: Proceedings of the Workshop on Human Evaluation of NLP Systems ,HumEval, pp 120–130, https://aclanthology.org/2021.humeval-1.14
Radford A, Wu J, Child R et al (2019) Language models are unsupervised multitask learners. OpenAI blog 1(8):9 (https://openai.com/blog)
Ouyang L, Wu J, Jiang X, et al (2022) Training language models to follow instructions with human feedback. pp 27730–27744, https://dl.acm.org/doi/10.5555/3600270.3602281
Achiam J, Adler S, Agarwal S, et al (2023) Gpt-4 technical report. arXiv preprint arXiv:2303.08774
Sun Y, Wang S, Li Y, et al (2020) Ernie 2.0: A continual pre-training framework for language understanding. In: Proceedings of the 34th AAAI Conference on Artificial Intelligence and 32nd Innovative Applications of Artificial Intelligence Conference and 10th AAAI Symposium on Educational

Advances in Artificial Intelligence, pp 8968–8975, https://doi.org/10.1609/aaai.v34i05.6428
Raffel C, Shazeer N, Roberts A et al (2020) Exploring the limits of transfer learning with a unified text-to-text transformer. J Mach Learn Res 21(140):1–67 (http://jmlr.org/papers/v21/20-074.html)
Chung HW, Hou L, Longpre S et al (2024) Scaling instruction-finetuned language models. J Mach Learn Res 25:1–53 (https://jmlr.org/papers/v25/23-0870.html)
Touvron H, Lavril T, Izacard G, et al (2023) Open and efficient foundation language models. Preprint at arXiv:1048550 2302. https://doi.org/10.48550/arXiv
Zeng A, Liu X, Du Z, et al (2022) Glm-130b: An open bilingual pre-trained model. arXiv preprint arXiv:2210.02414
Tay Y, Dehghani M, Tran VQ, et al (2022) Ul2: Unifying language learning paradigms. In: Proceedings of the 11th International Conference on Learning Representations,ICLR, https://iclr.cc/virtual/2023/poster/10748
Porsdam Mann S, Earp BD, Nyholm S et al (2023) Generative AI entails a credit-blame asymmetry. Nat Mach Intell 5(5):472–475. https://doi.org/10.1038/s42256-023-00653-1
Picazo-Sanchez P, Ortiz-Martin L (2024) Analysing the impact of chatgpt in research. Appl Intell 54(5):4172–4188. https://doi.org/10.1007/s10489-024-05298-0
Lin S, Hilton J, Evans O (2022) Truthfulqa: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association-for-Computational-Linguistics ,ACL, pp 3214–3252, https://doi.org/10.18653/v1/2022.acl-long.229
Strobelt H, Webson A, Sanh V et al (2022) Interactive and visual prompt engineering for ad-hoc task adaptation with large language models. IEEE Trans Visual Comput Graphics 29(1):1146–1156. https://doi.org/10.1109/TVCG.2022.3209479
Wang J, Xixu H, Hou W, et al (2023) On the robustness of chatgpt: An adversarial and out-of-distribution perspective. In: Proceedings of the ICLR 2023 Workshop on Trustworthy and Reliable Large-Scale Machine Learning Models,,ICLR, https://openreview.net/forum?id=uw6HSkgoM29
Shi J, Liu Y, Zhou P, et al (2023) Poster: Badgpt: Exploring security vulnerabilities of chatgpt via backdoor attacks to instructgpt. NDSS, https://dev.ndss-symposium.org/ndss2023/accepted-posters/
Fu Y, Peng H, Sabharwal A, et al (2022) Complexity-based prompting for multi-step reasoning. In: The Eleventh International Conference on Learning Representations, https://openreview.net/forum?id=yf1icZHC-l9
Zhang Z, Zhang A, Li M, et al(2023)Automatic chain of thought prompting in large language models. In: The Eleventh International Conference on Learning Representations, https://openreview.net/forum?id=5NTt8GFjUHkr
Shao Z, Gong Y, Shen Y, et al (2023) Synthetic prompting: Generating chain-of-thought demonstrations for large language models. In: International Conference on Machine Learning, PMLR, pp 30706–30775, https://dl.acm.org/doi/10.5555/3618408.3619681
Li H, Guo D, Fan W, et al (2023) Multi-step jailbreaking privacy attacks on chatgpt. In: The 2023 Conference on Empirical Methods in Natural Language Processing, https://doi.org/10.18653/v1/2023.findings-emnlp.272
Morris JX, Lifland E, Yoo JY, et al (2020) Textattack: A framework for adversarial attacks, data augmentation, and adversarial training in nlp. In: Proceedings of the 2020 conference on Empirical Methods in Natural Language Processing EMNLP, pp 119–126, https://doi.org/10.18653/v1/2020.emnlp-demos.16
Zeng G, Qi F, Zhou Q, et al (2021) Openattack: An open-source textual adversarial attack toolkit. In: Proceedings of the 59th Annual Meeting of the Association-for-Computational-Linguistics and11th International Joint Conference on Natural Language Processing and 6th Workshop on Representation Learning for NLP,ACL-IJCNLP, pp 363–371, https://doi.org/10.18653/v1/2021.acl-demo.43
Zhu K, Wang J, Zhou J et al (2024) Promptbench: A unified library for evaluation of large language models. J Mach Learning Res 25(254):1–22
Kojima T, Gu SS, Reid M, et al (2022) Large language models are zero-shot reasoners. Proceedings of the 36th Conference on Neural Information Processing Systems ,NeurIPS 35:22199–22213. https://dl.acm.org/doi/10.5555/3600270.3601883
Schick T, Schuetze H (2021) It's not just size that matters: Small language models are also few-shot learners. In: Proceedings of the 2021 conference the North-American-Chapter of the Association-for-Computational-Linguistics - Human Language Technologies ,NAACL-HLT, pp 2339–2352, https://doi.org/10.18653/v1/2021.naacl-main.185
Gao T, Fisch A, Chen D (2021) Making pre-trained language models better few-shot learners. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and 11th International Joint Conference on Natural Language Processing . ACL-IJCNLP, pp 3816–3830, https://doi.org/10.18653/v1/2021.acl-long.295
Zheng L, Chiang WL, Sheng Y et al (2024) Judging llm-as-a-judge with mt-bench and chatbot arena. Adv Neural Inform Process Syst 10(5555/3666122):3668142
Singh M, Cambronero J, Gulwani S, et al (2023) Codefusion: A pre-trained diffusion model for code generation. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing,EMNLP, pp 11697–11708, https://doi.org/10.18653/v1/2023.emnlp-main.716
De la Rosa J, Fernández A (2022) Zero-shot reading comprehension and reasoning for spanish with bertin gpt-j-6b. In: Proceedings of the Iberian Languages Evaluation Forum,IberLEF, https://ceur-ws.org/Vol-3202/
Black S, Biderman S, Hallahan E, et al (2022) Gpt-neox-20b: An open-source autoregressive language model. In: Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models, pp 95–136, https://doi.org/10.18653/v1/2022.bigscience-1.9
Le Scao T, Fan A, Akiki C, et al (2023) Bloom: A 176b-parameter open-access multilingual language model. arXiv preprint arXiv:2211.05100
Wang A, Singh A, Michael J, et al (2019) Glue: A multi-task benchmark and analysis platform for natural language understanding. In: Proceedings of the 7th International Conference on Learning Representations, EMNLP, p 353-355, https://doi.org/10.18653/v1/W18-5446
Wang B, Xu C, Wang S, et al (2021) Adversarial glue: A multi-task benchmark for robustness evaluation of language models. In: Proceedings of the 35th Conference on Neural Information Processing Systems Datasets and Benchmarks,NeurIPS, https://nips.cc/virtual/2021/47216
He X, Lyu L, Xu Q, et al (2021) Model extraction and adversarial transferability, your bert is vulnerable! In: Proceedings of the 2021 Conference of the North-American-Chapter of the Association-for-Computational-Linguistics - Human Language Technologies ,NAACL-HLT, pp 2006–2012, https://doi.org/10.18653/v1/2021.naacl-main.161
Wang J, Bao R, Zhang Z et al (2022) Rethinking textual adversarial defense for pre-trained language models. IEEE-ACM Trans Audio, Speech, and Language Process 30:2526–2540. https://doi.org/10.1109/TASLP.2022.3192097
Zang Y, Qi F, Yang C, et al (2020) Word-level textual adversarial attacking as combinatorial optimization. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics,ACL, pp 6066–6080, https://doi.org/10.18653/v1/2020.acl-main.540
Garg S, Ramakrishnan G (2020) Bae: Bert-based adversarial examples for text classification. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing EMNLP, 10.18653/v1/2020.emnlp-main.498
Ni M, Sun Z, Liu W (2024) Fraud's bargain attack: generating adversarial text samples via word manipulation process. IEEE Trans Knowl Data Eng 36(7):1–14. https://doi.org/10.1109/TKDE.2024.3349708
Liu S, Lu N, Chen C et al (2022) Efficient combinatorial optimization for word-level adversarial textual attack. IEEE-ACM Trans Audio, Speech Language Process 30:98–111. https://doi.org/10.1109/TASLP.2021.3130970

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**Xinxin Yue** is currently pursuing a master's degree at the School of Information Engineering, Henan University of Science and Technology, and the Henan International Joint Laboratory of Cyberspace Security Application in Luoyang, China. Her primary research interests lie in the security of LLMs.

**Zhiyong Zhang** (IEEE Senior Member, ACM Senior Member) received his Master and Ph.D. degrees in Computer Science from

Dalian University of Technology and Xidian University, P. R. China, respectively. He has been ever post-doctoral fellowship at School of Management, Xi'an Jiaotong University, China, and visiting professor of Computer Science Department of Iowa State University, US. Nowadays, he is Director of Henan International Joint Laboratory of Cyberspace Security Applications, Director of Henan Intelligent Manufacturing Big Data Development Innovation Laboratory, Executive Director of Information Science Department,Dean of College of Information Engineering,and full-time Henan Province Distinguished Professor at Henan University of Science and Technology, China. His research interests include cyberspace security and privacy computing, cyber-physical system and industrial internet security, as well as multimodel large model and social intelligence. Recent years, he has published over 150 scientific papers in IEEE TDSC, IEEE TCSS, IEEE TBD, etc, and edited 7 books in the above research fields, and also holds above 20 authorized patents. He is Chair of IEEE MMTC DRMIG, IEEE Systems, Man, Cybernetics Society Technical Committee on Soft Computing, World Federation on Soft Computing Young Researchers Committee, Committeeman of China National Audio, Video, Multimedia System and Device Standardization Technologies Committee. And also, he is editorial board member and associate editor of IEEE Access (IEEE), Human-centric Computing and Information Sciences (Springer), Multimedia Tools and Applications (Springer), Journal of Big Data (Springer), and leading guest editor or co-guest editor of Applied Soft Computing (Elsevier), Computer Journal (Oxford) and Future Generation Computer Systems (Elsevier). And also, he is Chair/Co-Chair and TPC Member for numerous international conferences/workshops on cyber security and privacy computing, big data and artificial intelligence.

**Junchang Jing** received the BS and MS in Mathematics and Information Science from Henan Normal University, Xinxiang, China, in 2015 and 2018, respectively, and the PhD degree in Information Engineering College, Henan University of Science and Technology, Luoyang, China, in 2023. He is currently an associate professor with the College of Computer and Information Engineering, Henan Normal University, Xinxiang, China. His research interests include network content security and computing, social situation analytics, multimodal large language model security and privacy.

**Weiguo Wang** received the Executive Master of Business Administration (EMBA) degree from emlyon business school. He is the Deputy General Manager and General Manager of Technical Department of the Henan Representative Office, New H3C Technologies Co., Ltd. He is also a member of Zhengzhou Big Data Standardization Technical Committee. He is engaged in the research in the fields of communication network, information security, cloud computing, big data and artificial intelligence.